Task 16: API Gateway and Load Balancing in **Spring Boot microservices**.

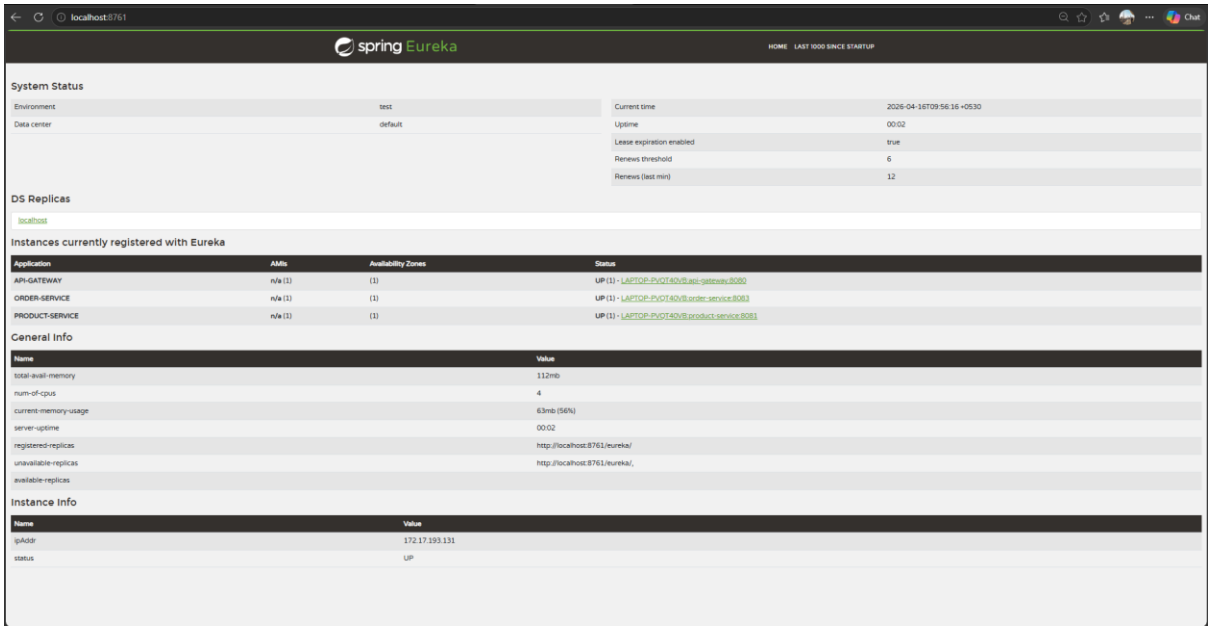
Aim

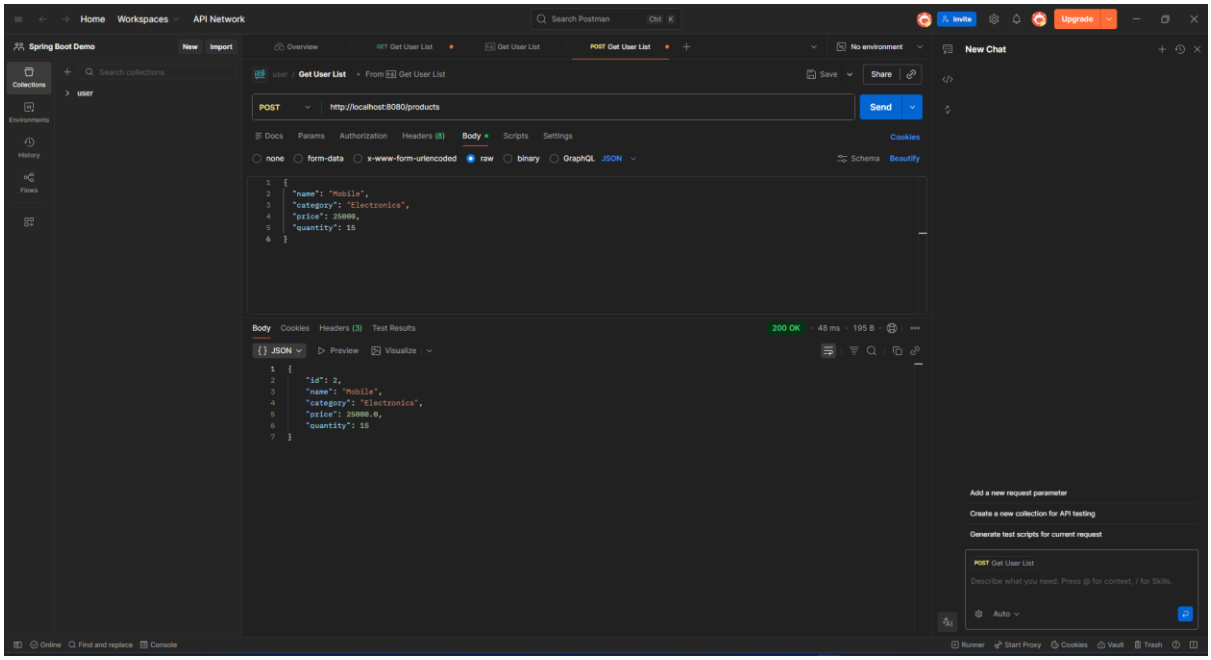
To configure an **API Gateway** in a microservices-based system for routing client requests to the appropriate microservices, and to implement **basic load balancing** so that requests are distributed across multiple instances of a service.

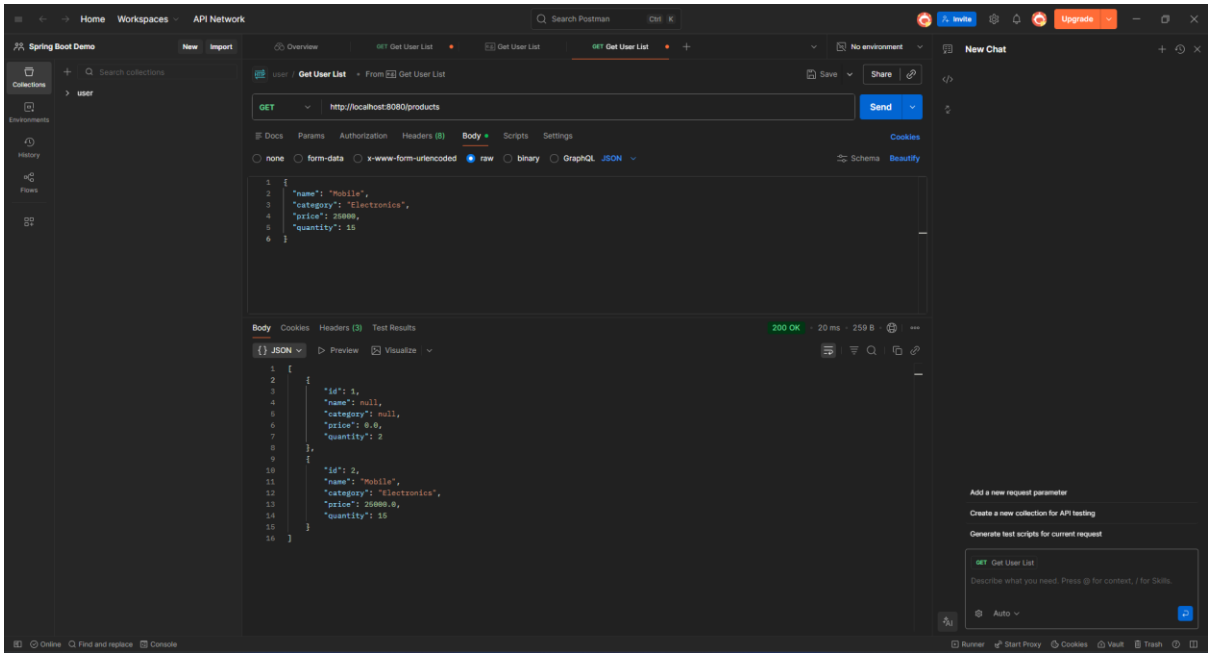
Algorithm

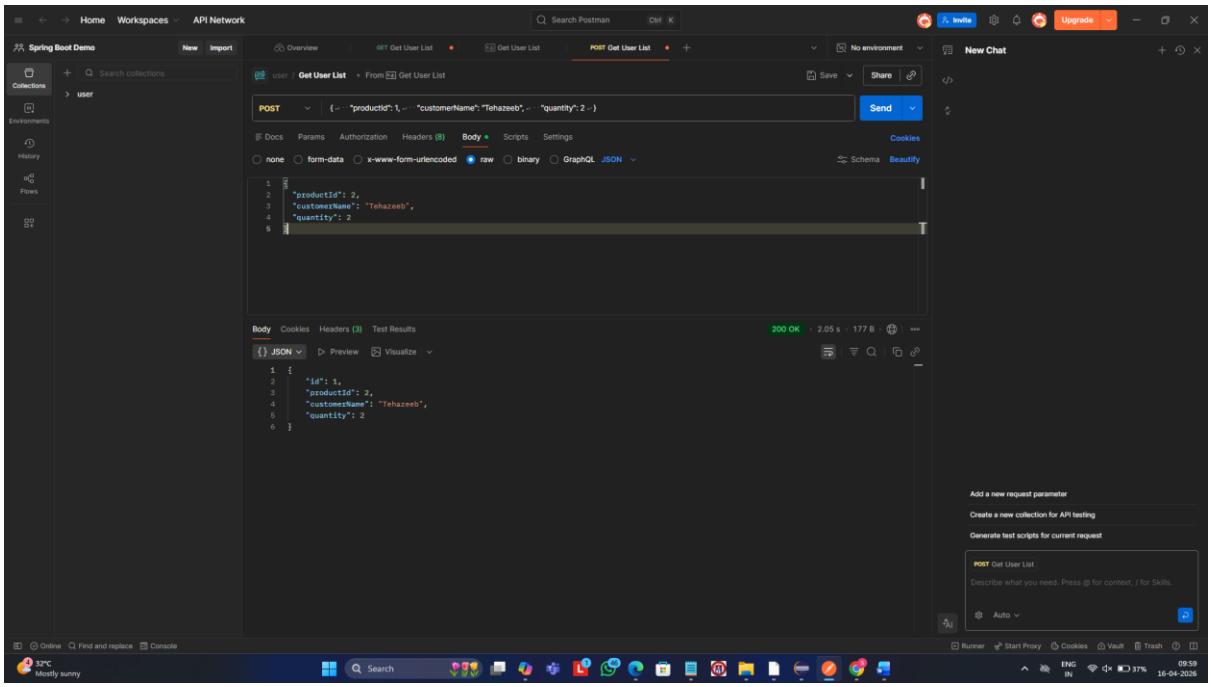
1. Start the microservices system.
2. Create an **API Gateway** using Spring Boot.
3. Add required dependencies for Gateway, Eureka Client, and Load Balancer.
4. Configure the API Gateway with a unique application name and server port.
5. Connect the API Gateway to the Eureka Server.
6. Define routing rules in the gateway configuration.
7. Map incoming client requests to the corresponding microservices.
8. Use service names instead of hard-coded URLs for routing.
9. Register multiple instances of the same microservice in Eureka.
10. Enable load balancing through the gateway.
11. Distribute incoming requests among available service instances.
12. Run Eureka Server, microservices, and API Gateway.
13. Send client requests to the API Gateway.
14. Verify that the requests are routed correctly.
15. Verify that repeated requests are balanced across service instances.
16. Stop the system.

Outputs:


localhost:8761
spring Eureka
HOME LAST 1000 SINCE STARTUP
System Status
Environment test
Data center default
Current time 2026-04-16T09:56:16 +0530
Uptime 00:02
Lease expiration enabled true
Renews threshold 6
Renews (last min) 12
DS Replicas
localhost
Instances currently registered with Eureka
Application AMIs Availability Zones Status
API-GATEWAY n/a (1) (1) UP (1) - LAPTOP-PVQT40VB:api-gateway:8080
ORDER-SERVICE n/a (1) (1) UP (1) - LAPTOP-PVQT40VB:order-service:8083
PRODUCT-SERVICE n/a (1) (1) UP (1) - LAPTOP-PVQT40VB:product-service:8081
General Info
Name Value
total-avail-memory 112mb
num-of-cpus 4
current-memory-usage 63mb (56%)
server-uptime 00:02
registered-replicas http://localhost:8761/eureka/
unavailable-replicas http://localhost:8761/eureka/,
available-replicas
Instance Info
Name Value
ipAddr 172.17.193.131
status UP


Home Workspaces API Network
Search Postman
Invite
Upgrade
Spring Boot Demo
New Import
Collections
Environments
History
Flows
Search collections
user
Overview
GET Get User List
Get User List
POST Get User List
No environment
New Chat
user / Get User List
From Get User List
Save
Share
POST
http://localhost:8080/products
Send
Docs Params Authorization Headers (8) Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL JSON
Schema Beautify
{
"name": "Mobile",
"category": "Electronics",
"price": 25000,
"quantity": 15
}
Body Cookies Headers (3) Test Results
200 OK 48 ms 195 B
JSON Preview Visualize
{
"id": 2,
"name": "Mobile",
"category": "Electronics",
"price": 25000.0,
"quantity": 15
}
Add a new request parameter
Create a new collection for API testing
Generate test scripts for current request
POST Get User List
Describe what you need. Press @ for context, / for Skills.
Auto
Online Find and replace Console
Runner Start Proxy Cookies Vault Trash





Result:

To configure an **API Gateway** in a microservices-based system for routing client requests to the appropriate microservices, and to implement **basic load balancing** so that requests are distributed across multiple instances of a service is executed successfully.